

The EM Algorithm

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A Simple Measurement Problem

- Imagine that a factory produces the same component using two different machines.
- Let's say there is a common measurable quantity of the produce of both the machines, say the diameter of a component.
- Suppose:
 - ① Machine 1 tends to produce components around 20 mm.
 - ② Machine 2 tends to produce components around 25 mm.
- We collect measurements from the production line:

20.3, 24.8, 25.2, 19.7, 21.1, 24.5, 20.8, 25.4, ...

- But there is a problem.

We did not record which machine produced each component

What Would We Like to Know?

- We would like to recover information about the two machines.
- For example:

μ_1 = average output of Machine 1

μ_2 = average output of Machine 2

σ_1 = variability of Machine 1

σ_2 = variability of Machine 2

- We might also want to know:

π_1 = fraction of production from Machine 1

π_2 = fraction of production from Machine 2

- So from the measurements alone, we want to reconstruct the characteristics of the two machines.

Machine Information Were Available VS Labels Are Missing

- Suppose the production records had been maintained properly:

Measurement	20.3	24.8	25.2	19.7	21.1	24.5	20.8	25.4
Machine	1	2	2	1	1	2	1	2

- Then the problem would be easy. We could separate the measurements and calculate the mean and variance of each group.
- In reality, all we have is
20.3, 24.8, 25.2, 19.7, 21.1, 24.5, 20.8, 25.4, ...
- We don't know which machine produced which measurement. This missing information is the central difficulty.

Why Can't We Just Separate the Data?

- Looking at the data, we might say:
Values near 20 probably came from Machine 1, and values near 25 probably came from Machine 2.
- But consider $x = 22.8$. Which machine produced it?
- Or suppose the two machines have more variability. There is **no sharp boundary** separating the two populations.
- So assigning $x = 22.8$ completely to one machine may be unreasonable.

A Better Question

- Instead of asking
Which machine produced this observation?
- we ask
How likely is each machine to have produced this observation?

A Better Question

- For example, for $x = 22.8$, we might obtain

	Machine 1	Machine 2
$x = 22.8$	0.7	0.3

- We don't have to make a hard decision. We can say:

70% associated with Machine 1, 30% associated with Machine 2.

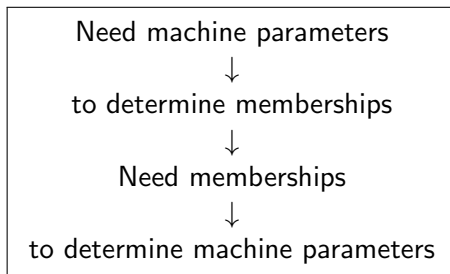
- How are these numbers of any help?
- These numbers will become **weights** when estimating the machine parameters.
- All well and good still we only have raw data, how do we assign any likelihood this data?
- this will need us to know more about the machines, isn't it !!

Now We Have a Circular Problem

- It is an interesting and peculiar difficulty.
- To decide how likely each machine is to have produced x , we need to know the characteristics of the machines.
- For example, we need to know

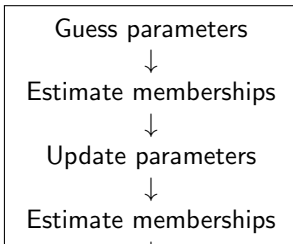
$$\mu_1, \sigma_1, \mu_2, \sigma_2.$$

- But to estimate those characteristics, we need to know which measurements came from each machine! So we have a circular problem:



Breaking the Circle

- This is exactly the situation where EM becomes useful.
- We can break this circularity by **alternating between the two tasks**.
- Start with some rough guesses for the machine parameters. Then:
 - 1 Use our current guesses to estimate how strongly each observation belongs to each machine.
 - 2 Use these estimated memberships to improve the machine parameters.
- Then use the improved parameters to estimate the memberships again. So:



A Very Simple Analogy

- Think of two teachers trying to determine which of two courses a set of students most likely belongs to.
- Initially, they have rough ideas about the typical students in each course.
- They then:
 - 1 **Estimate:** Based on the current descriptions, how strongly does each student resemble each course?
 - 2 **Update:** Based on these assignments, revise the description of each course.
 - 3 **Repeat.**
- The descriptions become better, which makes the assignments better, which makes the descriptions better again.
- The intuition behind EM:

better assignments	$\longleftrightarrow$	better parameters
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Where Does the Name EM Come From?

- The two alternating steps have names.
 - ① **E – Expectation:** Estimate the **expected/soft membership** of each observation.

Which source is responsible, and by how much?

- ② **M – Maximization:** Using those memberships, find the parameters that best describe the data.

What parameters best fit these weighted observations?

E-step $\longrightarrow$ M-step $\longrightarrow$ E-step $\longrightarrow \dots$